

COLE H. ROTTENBERG

(407) 961-0600 | cole.rottensberg@gmail.com | github.com/colerottensberg

EDUCATION

University of Florida Herbert Wertheim College of Engineering
Bachelor of Science in Computer Engineering

Gainesville, FL
Dec 2024

Relevant Coursework: Operating Systems, Fundamentals of Machine Learning, Applied Machine Learning, Introduction to Competitive Programming, Digital Design, Microprocessor Applications, Data Structures and Algorithms

EXPERIENCE

AlgoGators Investment Fund

Chief Technology Officer | Founder

Gainesville, FL
Aug 2023 – Present

- Designing an asynchronous data pipeline that ingests real-time market data across 50+ futures instruments and 100+ contracts.
- Implementing a centralized system combining divergent and convergent strategies with risk and dynamic optimization to generate positions.
- Developing and deploying scalable data stores using parquet buckets to manage both real-time and historical market data from multiple exchanges enabling efficient storage, retrieval, and processing of data.
- Deploying a fully reproducible systematic trading environment using Ubuntu-based Linux with Nix and Nix flakes, ensuring seamless updates and system reliability across multiple deployments.
- Automating the recalculation of target positions, triggered by new incoming market data, resulting in a system that continuously optimizes based on changing conditions without manual intervention.
- Integrating Poetry as the fund's package and project management system, vastly improving development cycle times and ensuring consistency in software builds across environments.
- Leading a team of a dozen engineers and quants in developing and refining trading algorithms, optimizing execution efficiency, and scaling the system to support an expanding portfolio of assets and strategies.

Blue Vigil

Software Engineering Intern

Sterling, VA
Jun 2023 – Aug 2024

- Developed robust communication protocols over RS-485 in electrically noisy environments, integrating Reed-Solomon error-correction reducing data corruption rates from 90% to 40%.
- Standardized messaging protocols by transforming our custom messaging format into a company-wide library.
- Implemented continuous integration (CI/CD) testing and deployment workflows to ensure the robustness of the library.
- Utilized Arduino to index and manage the package for efficient deployment across devices.
- Developed a multi-flight log analysis software program for interpreting critical flight and power data within a drone system, automatically parsing and analyzing flight logs and a control summary for each flight using PyQt, NumPy, Matplotlib, and pandas.

PROJECTS AND PROFESSIONAL DEVELOPMENT

KneeKare Pro

Embedded Systems Engineer | Backend Engineer

Gainesville, FL
Jan 2024 – Present

- Engineering robust firmware in C++ for an ESP32-based wearable knee flexion monitor, enabling real-time streaming of angle and velocity data via Bluetooth Low Energy (BLE), with a latency of less than 50ms, supporting clinicians in assessing rehabilitation progress post-injury.
- Designing and implementing a scalable backend using Flask, interfaced with InfluxDB, to handle patient time-series data, enabling the detection of rehabilitation trends across multiple time horizons.
- Developing a mobile application to stream data from a Bluetooth Low-Energy(BLE) knee monitor, enabling real-time analysis of rehabilitation metrics to track patient recovery progress efficiently.

Fundamentals of Machine Learning Competition

AI Researcher

Gainesville, FL
Mar – Apr 2024

- Led the end-to-end development of a deep learning model for detecting American Sign Language (ASL) gestures, leveraging a Convolutional Neural Network (CNN) inspired by the MobileNet architecture.
- Utilized TensorFlow's Keras API for efficient model construction and training, achieving an accuracy of 96% in sign recognition. Accelerated computational processes were conducted on two Nvidia A100 GPUs, significantly reducing model training time by 80%.

ADDITIONAL INFORMATION

- **Skills:** C, C++, Linux, Python, SQL, TensorFlow, scikit-learn, NumPy, pandas, Rust, Nix, NixOS, Git/GitHub, Jupyter
- **Memberships:** Florida Cryptocurrency and Alternative Assets Club, Quantitative Finance Track
- **High School:** Varsity Lacrosse and Swimming